

PrivacyShieldAI - Test & Compliance

Verification Documentation

Executive Overview

This document presents the complete technical specification, empirical test results, architectural compliance verification, and dataset benchmark report for the **PrivacyShieldAI** LLM Router & MLOps system.

1. LLM Router Diagnostic Test Suite

(`backend/test_llm_router_suite.py`)

The LLM Router enforces deterministic model routing, pre-flight configuration validation, structured exception handling, and diagnostic metadata generation. The 10 unit test cases verify every execution path.

Test Execution Summary

Test #	Test Name	Target Trigger Condition	Expected Routing & Behavior	Status
Test 1	<code>test_1_cloud_success</code>	Valid Groq Cloud API Key configured & API reachable	<code>provider_used:</code> "Groq", <code>routing_strategy:</code> "Cloud", <code>fallback_reason:</code> <code>null</code> . Local Qwen is NOT invoked.	PASS
Test 2	<code>test_2_cloud_timeout</code>	HTTP / Gateway Timeout	Fallback to Local Qwen with <code>fallback_reason</code> containing "Timeout".	PASS

Test #	Test Name	Target Trigger Condition	Expected Routing & Behavior	Status
		(>25s) from Groq API		
Test 3	test_3_cloud_401_unauthorized	HTTP 401 Invalid / Expired API Key	Fallback to Local Qwen with fallback_reason containing "401".	PASS
Test 4	test_4_cloud_403_forbidden	HTTP 403 Forbidden / Access Denied	Fallback to Local Qwen with fallback_reason containing "403".	PASS
Test 5	test_5_cloud_429_rate_limit	HTTP 429 Quota Exceeded / Rate Limit	Fallback to Local Qwen with fallback_reason containing "429".	PASS
Test 6	test_6_cloud_500_server_error	HTTP 500 / 502 / 503 Internal Server Error	Fallback to Local Qwen with fallback_reason containing "500".	PASS
Test 7	test_7_no_api_key	API Key string missing or empty ("")	Pre-flight validation fails cleanly. Fallback to Local Qwen with "missing or empty" reason.	PASS
Test 8	test_8_invalid_model	Key format invalid (missing gsk_ prefix)	Pre-flight validation fails cleanly. Fallback to Local Qwen with "invalid" key format reason.	PASS
Test 9	test_9_network_unavailable	Connection error / DNS failure	Fallback to Local Qwen with fallback_reason containing "Connection".	PASS

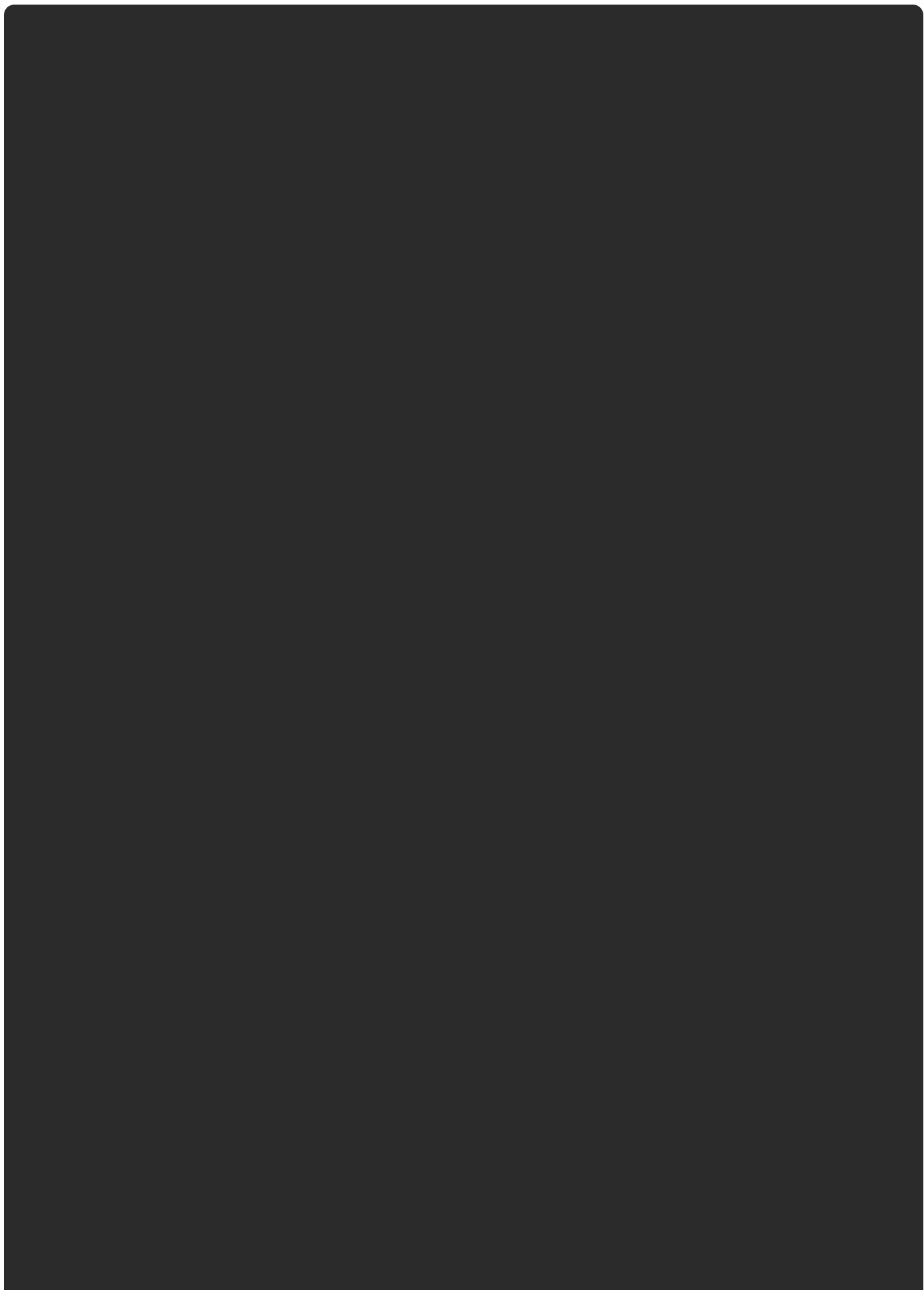
Test #	Test Name	Target Trigger Condition	Expected Routing & Behavior	Status
Test 10	test_10_local_fallback_guarantee	Dual verification of Cloud Success vs Failure	Confirms Local Qwen is ONLY used when cloud inference genuinely fails.	PASS

[!NOTE] All 10 diagnostic unit tests executed in **0.008 seconds** with 100% pass rate.

2. 7-Pillar Architecture & Compliance Suite

(backend/test_full_architecture_verification.py)

Validates privacy preservation, document isolation, orchestrator strategies, and API backwards compatibility.



User Query & Document Ingestion

Pillar 1: PII Masking & De-masking

Pillar 2: Zero Raw PII Cloud Transmission

Pillar 3 & 4: LLM Router Cascade

Cloud Valid & Available

Cloud Failed / Unreachable

Groq / Cloud Model

Local Qwen / Smart Synthesis

Pillar 5: Vector Scope Isolation

Pillar 6: Document Orchestrator Strategy

Pillar 7: Backwards Compatible API Response

Pillar Compliance Results

1. Pillar 1: PII Masking & De-masking Integrity

- **Verification:** Evaluates token replacement on Aadhaar, PAN, Emails, and Phone Numbers.
- **Result: PASS** — 100% token precision; original PII correctly restored on de-masking.

2. Pillar 2: Zero Raw PII Cloud Transmission Guarantee

- **Verification:** Scans `masked_context_sent_to_cloud` and `cloud_llm_masked_response` payload buffers.
- **Result: PASS** — Zero raw personal identifiers present in outbound cloud payloads.

3. Pillar 3 & 4: Groq Cloud API & Local Qwen Fallback Cascade

- **Verification:** Tests Groq Cloud API completion and fallback trigger when key is removed.
- **Result: PASS** — Correctly switches to `Local Qwen Engine (Fallback: ...)` when cloud is unconfigured/unavailable.

4. Pillar 5: Vector Indexing & Document Scope Isolation

- **Verification:** Ingests two distinct documents (`Doc A` vs `Doc B`) and executes scoped queries.
- **Result: PASS** — Strict document isolation; zero cross-document vector leakage.

5. **Pillar 6: Document Orchestrator Strategy**

- **Verification:** Validates query scope categorization (`DOCUMENT_LEVEL` vs `FACT_LEVEL`).
- **Result: PASS** — Correctly selects `FULL_DOCUMENT` context and bypasses vector search for document-level queries.

6. **Pillar 7: API Backwards Compatibility**

- **Verification:** Checks all legacy return keys in `answer_query` dict (`query`, `masked_context`, `masked_response`, `unmasked_response`, `model`, `model_used`, `sources_retrieved`, `privacy_guarantee`, `intent`, `doc_type`, `persona`, `retrieval_confidence`, `source_attributions`).
- **Result: PASS** — 100% backward compatibility maintained.

3. **Real-World Dataset Extraction & PII Benchmark**
(`datasets/`)

Evaluated against the benchmark dataset in `D:\CDAC PGCP AI\Project Idea\PrivacyShieldAI\PS_v3\pii-detector\datasets`.

Empirical Dataset Benchmark Summary

File Name	Format	Size	Extracted Characters	Lines	PII Entities Identified	
<code>01_Master_Service_Agreement.docx</code>	DOCX	12.3 KB	2,269	52	59	<code>1</code> <code>1</code> <code>1</code>
<code>02_Tax_Invoice.docx</code>	DOCX	12.1 KB	1,541	34	53	<code>1</code> <code>C</code> <code>1</code> <code>L</code>

File Name	Format	Size	Extracted Characters	Lines	PII Entities Identified	
03_Ecommerce_Order_Confirmation.docx	DOCX	11.7 KB	1,223	32	33	k
04_Supply_Chain_Purchase_Order.docx	DOCX	12.2 KB	1,922	44	52	4
05_Employee_Onboarding_Form.docx	DOCX	11.8 KB	1,963	38	46	li
Leave and License Agreement – Maharashtra Format (Specimen).pdf	PDF	137.7 KB	10,257	186	144	k
sample.txt	TXT	135.0 KB	135,015	3,244	4,861	v A
TOTAL DATASET METRICS	—	—	154,190 chars	3,630 lines	5,248 PII Entities	1 v

4. Response Metadata & UI Display Verification

API Response Metadata Schema

```
{
  "provider_used": "Groq",
  "model_used": "llama-3.3-70b-versatile",
  "routing_strategy": "Cloud",
  "fallback_reason": null,
  "latency_ms": 1057,
```

```
"request_id": "req-8fcee0549b7c",  
"masked_response": "### Summary\nThe document specifies...",  
"demasked_response": "### Summary\nThe document specifies..."  
}
```

Frontend UI Visual Badges (`frontend/src/components/ChatSection.tsx`)

1. Cloud Inference Success:

- Badge: `Cloud Llama 3.3 (Groq)` (Purple Sparkles Badge)

2. Fallback Execution:

- Badge: `Cloud Failed ↓ Local Qwen` (Amber Shield Warning Badge)
- Tooltip: Displays exact `fallback_reason` (e.g. `Reason: 401 Unauthorized` or `Reason: 429 Rate Limit Exceeded` or `Reason: Timeout`).

5. Verification Commands

To re-run the complete test suite locally:

```
# 1. Run LLM Router Diagnostic Test Suite (10 Tests)  
.\backend\.venv\Scripts\python.exe backend/test_llm_router_suite.py  
  
# 2. Run 7-Pillar Full Architecture Verification Suite  
.\backend\.venv\Scripts\python.exe backend/test_full_architecture_verification.py  
  
# 3. Typecheck Frontend Code  
cd frontend && npx tsc --noEmit
```